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Magnetic polarity stratigraphy of the Tertiary igneous rocks of Skye, Scotland

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SUMMARY

Palaeomagnetic results are given for 115 sites, mainly from the central igneous intrusive complex, and also radiometric dates by the ^{40}Ar–^{39}Ar step-heating method. These are combined with published dates and palaeomagnetic results for the lavas and regional dyke swarms with the aim of establishing the magnetic polarity stratigraphy.

Igneous activity spanned at least the approximate interval 61–54 Ma, and a polarity sequence of at last R–N–R–N–R, but could span seven polarity intervals, R–N–R–N–R–N–R, depending on the age of the Loch Ainort Epigranite. However, these polarities cannot be identified with specific chrons of either the Harland *et al.* (1982) or Berggren *et al.* (1985) polarity time-scales.

The Skye igneous activity, therefore, spans more polarity intervals than in Mull, Ardnamurchan or Rhum and this is consistent with the radiometric ages.

Interpretations of the results for the Coire Uaigneich and related granites and for the Loch Ainort Epigranite present problems, and partial solutions are offered.

Key words: igneous, overprinting, palaeomagnetism, radiometric dating, Skye, Tertiary.

INTRODUCTION AND GEOLOGY

Palaeomagnetic sampling has been carried out in Skye, primarily of the central igneous intrusive complex, to complement earlier studies of the lavas (Wilson, Dagley & Ade-Hall 1972a) and dykes (Wilson *et al.* 1974; Wilson, Hall & Dagley 1982). This paper describes the new results and uses all the results to try to establish a magnetic polarity stratigraphy for the Tertiary igneous rocks of Skye, and to help define the timing and duration of igneous activity.

Igneous activity commenced with the outpouring of vast quantities of basic lavas which still cover most of the island [Harker (1904); see Bell & Harris (1986) for a summary]. The lavas have been divided into five groups of which the order of formation has been partly deduced (Anderson & Dunham 1966). The igneous intrusive complex developed within the lava field; with the focus of activity migrating, producing four centres within the complex. The first centre was the Cuillins (CC), predominantly of basic and ultrabasic rocks. The centre of activity then moved to the east, thence to the NE and finally to the SE, forming in turn the Srath na Creitheach (SNCC), Western Red Hills (WRHC) and Eastern Red Hills (ERHC) Centres; these three centres are composed predominantly of acid rocks. Bell (1976) has placed the many component units of the central intrusive complex in a tentative sequence of events (Table 1).

Minor intrusions are common, particularly basic dykes which mostly belong to regional swarms. Cutting relationships show that these dykes were intruded throughout the period of Tertiary igneous activity in Skye, and Speight (1972; Speight *et al.* 1982) has mapped the axes of the main swarms and their percentage dilations.

Oxygen isotope studies on the Tertiary igneous rocks of Skye by Taylor & Forester (1971) and Forester & Taylor (1977) have shown that many of them are depleted in ^{18}O as a result of interaction with a meteoric hydrothermal system driven by the intrusions forming the central complex. Frequently the rocks have been mineralogically altered by the hydrothermal system.

Ferry (1985a, b) gives a detailed account of the alteration but the more important changes can be summarized as follows.

In the granites: amphibole, clinopyroxene and biotite are partially altered to chlorite; fayalite is frequently replaced by brown montmorillinite; alkali feldspar and plagioclase become turbid and with progressive alteration plagioclase is altered to epidote, calcite and muscovite; titaniferrous

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Table 1. Summary of Skye stratigraphy and magnetic polarities (based on Bell 1976, table 1, p. 252).

Focus of Activity	Event No. After Bell (1976)	Intrusive events	Abbreviation	Event Polarity	Sampling site number(prefixed U) and polarity	Radiometric ages and other comments
		DYKES (i) CUTTING C-A SILLS (48)		R	95R;96R.	
		(ii) CUTTTING BEINN NA CAILLICH G.(47)		R	24R.	
		(iii) CUTTING BEINN AN DUBHAICH (44)		R	100R;101R.	
		(iv) PRE BEINN AN DUBHAICH (44)		R	7R.	
	48	COMPOSITE AND ACID SILLS	CAS	R	8R;94R.	55.0±0.8/55.3±0.9[8]
	47	BEINN NA CAILLICH EPIGRANITE	BNCE"	R	72R.	
	46	BEINN NA CAILLICH VENT	(VA)	-	not sampled	
EASTERN	45	KILCHRIST HYBRIDS	KH	R	85R;86R.	54.0±1.4[8]
RED HILLS	49	CREAG STROLLAMUS EPIGRANITE	CSE	?	10?	)related intru-
CENTRE	44	BEINN AN DUBHAICH EPIGRANITE	BADE	R	6R;117R;118R.	53.5±0.8[6])sions Stewart
	43	ALLT FEARNA EPIGRANITE	AFE	?	11?;25?.	)(1965) Bell &
	42	BEINN NA CRO EPIGRANITE	BNCE'	?	64?	)Harris(1986)
	41	GLAS BEINN MHOR EPIGRANITE	GBME	R	9R.	
	40	BROADFORD GABBRO	BG	R	12R.	
	39	BEINN NA CRO GABBRO	BNCG	R	23R.	
	38	KILCHRIST VENT	(VA)	-	not sampled	
	37	DYKES		N	? 002N.	
	36	MARSCO EPIGRANITE	ME	N	28N;69N;89N.	
	35	MEALL BUIDHE EPIGRANITE	MBE	-	not sampled	
	34	MARSCOITE SUITE	MS	N	60N;70N;841;87N.	
	33	SOUTHERN PORPHYRITIC EPIGRANITE	SPE	N	83N;88N.	
	32	NORTHERN PORPHYRITIC FELSITE	NPF	N	26N.	
WESTERN	31	GLEN SLIGACHAN EPIGRANITE	SE	N	15N.	
RED HILLS	30?	LOCH AINORT EPIGRANITE	LAE	R	1R;62R;93R;97R.	58.7±1.7[6]
CENTRE	29	BEINN DEARG MHOR EPIGRANITE	BDME	N	13?;63N.	
	28	EAS MOR EPIGRANITE	EME	N	14N	
	27	MAOL NA GAINMHICH EPIGRANITE	MNGE	N	61N.	
	26	GLAMAIG EPIGRANITE	GE	N	27N.	
	25	MARSCO SUMMIT GABBRO	MSG	-	not sampled	
	24	BELIG VENT (?and MEALL A'MHAOIL VENT)	(VA)	-	not sampled	
	23	DYKES				
SRATH NA	22	BLA BHEINN (BLAVEN) EPIGRANITE	BE	-	not sampled	
CREITHEACH	21	MEALL DEARG EPIGRANITE	MDE	R	20R.	
CENTRE	20	RUADH STAC EPIGRANITE	RSE	R	29?;91R.	
	19	LOCH NA CREITHEACH VENT	(VA)	-	not sampled	
	18	DYKES		N	34N.	
	17	CONE SHEETS				
		(i) CUTTING CUG (16)				
		at COIRE UAIGNEICH		N	32N;33N;116N;113N.	
		at CAMASUNARY		R	73R;82R.	
		(ii) CUTTING GABBRO				
		at N SIDE of GARBH-BHEINN		R	66N;67N;123R	
		on W SLOPES of BLA BHEINN		R	129R.	
		at SGURR NA STRI		R	76R;78R;80R.	
	16	COIRE UAIGNEICH and other GRANITES	CU			
		(i) COIRE UIAGNEICH		N	31N;112N;114N;115N.	59.3±0.7[7]
		(ii) CAMASUNARY		N	74N;81I.	
		(iii) S.FLANK OF BELIG		R	71R.	differs petrographically from i&ii Bell (1976) + this paper.
	15	INTRUSIVE THOLEIITES - OUTER COMPLEX		-	not sampled	
	14	-MAIN RIDGE COMPLEX		-	not sampled	
	13	*DRUIM NAN RAMH RING EUCRITE*	DNRRE	R	16R;18R	
	12	AGGLOMERATES AND EXPLOSION BRECCIAS	(VA)		not sampled	
CUILLIN	11	DYKES		-	not sampled	
CENTRE	10	?GARS-BHEINN ULTRABASIC SILL		-	not sampled	
	9	GABBRO SERIES	G			
		GLEN SLIGACHAN		R	92R.	
		MEALL DEARG		R	30R.	
		MEALLAN DEARG		R	19R.	
	8	EUCRITE SERIES	ES			
		N.SIDE of GARBH BHEINN		R	65N;68R;119R;120R;121R;122R.	
		W.SLOPES of BLA BHEINN		R	124R;125R;126R;127R;128R.	
		SGURR NA STRI		R	75R;77R;79R.	
		LOCH SCAVAIG		R	104R;105R.	
	7	SGURR DUBH PERIDOTITE SERIES	P	R	108R;109R.	
	6	ALLIVALITE SERIES	A"	-	not sampled	
	5	BORDER GROUP(including White Allivalite)[4]	BZ	R	106R;107R.	
	4	CONE SHEETS		R	21R;22R.	
	3	DYKES		-	not sampled	
	2	RING EUCRITE AND OUTER GABBROS				
		(i) GLEN SLIGACHAN	OG	R	90N.	
		(ii) COIRE NA CREICHE	OG	R	102R;103R.	
		(iii) GARS-BHEINN GABBRO[5]	GBG	R	110N;111R.	
	1	EARLY GRANITES(occurring as fragments in later vents)		-	not sampled	

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Table 1. (*continued*)

Focus of activity	Event No.	Intrusive events	Abbreviation	Event Polarity	Sampling site number and Polarity	Radiometric ages and other comments
PLATEAU LAVAS				R		
		OSDALE GROUP[1]		R	M007 - M026 R	
		BRACADALE GROUP[1]		R	L001 - L006 R	
		BEINN TOTAIG GROUP		R	U051 - U056 R	
		RAMASCAIG GROUP[1]		R	J055 - J077 R	
				R	K003 - K022 R	
		BEINN EDRA GROUP[1]		R	N001 - N039 R	
		PRESHAL MHOR TYPE		R	U003 - U004 R	
		FAIRY BRIDGE TYPE		R	U005 - U005A R	
		UNGROUPED		R	U041	
REGIONAL DYKES OF NW TREND (probably intruded throughout period of activity)		VATERNISH SWARM[2]		N+R	I000 - I081 N+R	
		SLEAT SWARM[3]		N+R	H001 - H382 N+R	
SILLS		CUTTING PLATEAU LAVAS		R	U017 R	

1. Wilson et al. 1972a
2. Wilson et al. 1974
3. Wilson et al. 1982
4. Bell 1976
5. Bevan & Hutchison 1984
6. Dickin 1981
7. Dickin & Exley 1981
8. THIS PAPER

magnetite has been partially converted to ilmenite and Ti-poor magnetite.

In the gabbros and dolerites: olivine is replaced by talc and magnetite and occasionally by chlorite; pyroxene is altered to green calcic amphibole; Ti magnetite is partially converted to ilmenite and Ti-poor magnetite.

Figs 1, 2 and 3 show the principal geological features [for summaries see Bell (1976) and Bell & Harris (1986)] and all the sampling locations.

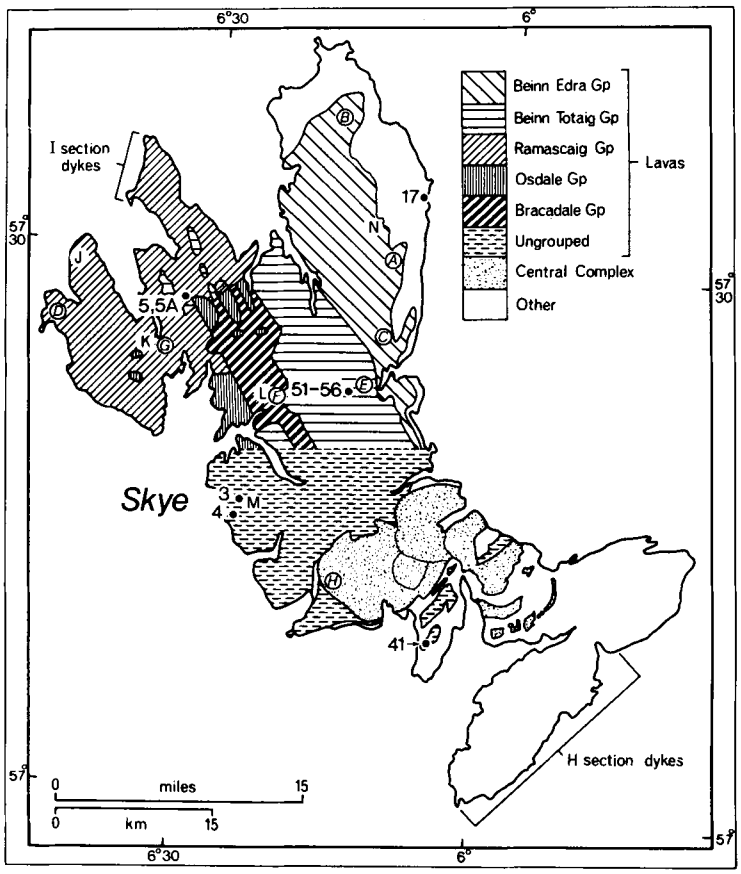


Figure 1. Sketch map of the geology of Skye showing sampling sites outside the central intrusive complex (after Anderson & Dunham 1966). *Numbers* refer to this study [Table 1 and Appendix, Table A2(i), deposited in RAS Library]; *uncircled* letters to lava and dyke sequences of Wilson *et al.* (1972a, 1974, 1982) [Table 1 and Appendix, Table A2(iii)]; *circled* letters to lava sequences of Khan (1960) [Appendix, Table A2(iii)].

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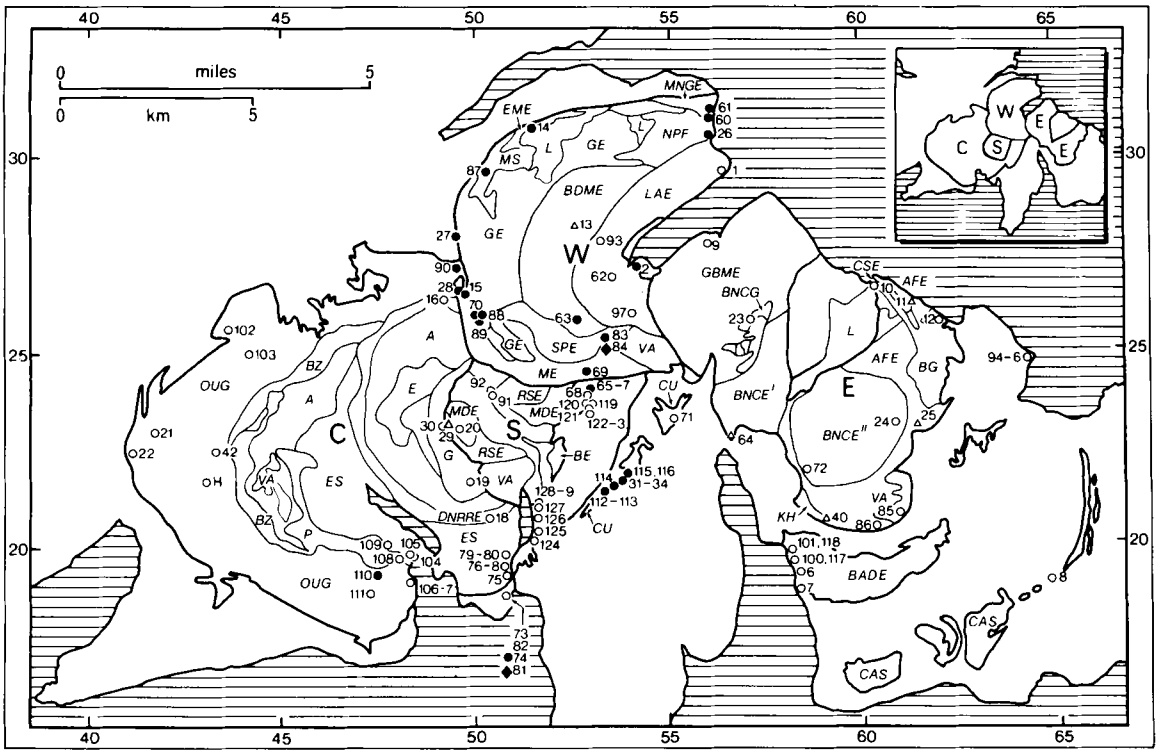


Figure 2. Sketch map of the major units of the central igneous intrusive complex showing sampling sites (after Bell 1976). The coordinates refer to the kilometre squares of sheet NG of the National Grid References System. Full circles = normal polarity; open circles = reversed polarity; full diamonds = intermediate polarity; open triangles = undefined.

SAMPLING AND MEASURING

Sampling followed our usual procedure. Briefly, 25 mm-diameter cores were drilled, four or more per site, and oriented independently by sun-sighting or by using geographical bearings. Sites were selected to include as many of the geological units as possible. Some units were sampled several times to resolve problems of overprinting or other complications; for example certain units were sampled at several widely spaced sites and traverses were made near geological boundaries. A grid reference and description for each sampling site is given in the Appendix, Table A1 (deposited in Royal Astronomical Society Library as 'Add. m/s. 191').

The freshest possible rock also was sampled from units in several parts of the igneous succession for dating by the ^{40}Ar–^{39}Ar step-heating method.

In the laboratory 25 mm-long specimens were demagnetized by alternating fields in steps (usually of 5 mT), and the site mean directions were deduced using the method of Dagley & Ade-Hall (1970). Additional specimens from selected sites were thermally demagnetized.

For most sites average bulk magnetic susceptibility and Curie temperature were measured.

Many samples were examined petrographically and radiometric age determinations were made as described below.

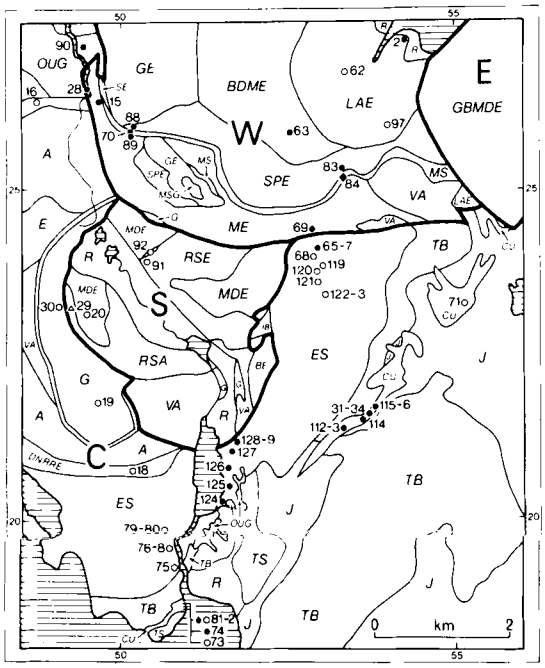


Figure 3. Enlarged part of Fig. 2 to show areas affected by overprinting. Symbols and coordinate grid as for Fig. 2.

PALAEOMAGNETIC RESULTS

In general, most of the 115 sites gave a clearly defined mean direction ($\alpha_{95} < 10°$) but a few may have been affected by lightning and some, mainly granites, were too weakly magnetized or unstable to give a clear direction. These few have Q ratios less than 0.5 whereas the average value is 4.0. The majority of samples have Curie temperatures in excess of 500 °C, the average for those greater than 450 °C being 558 °C. With only two exceptions (sites 8 and 84) the Curie

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points less than 500 °C are associated with reversely magnetized dykes and lavas.

The stable directions obtained after demagnetization are considered to be primary directions; however, the directions obtained for different sites in the same unit do not always agree within the uncertainties. This could be due to a combination of slow differential cooling and secular variation, to localized remagnetization or to minor tectonic movement, perhaps associated with the emplacement of later intrusions. As a group, the reversed directions are shallower, and more scattered than the normal ones (Fig. 4).

The mean directions are given in the Appendix, Table A2 (deposited with Table A1 in RAS Library) for all sites, including those published previously; polarities are shown in Figs 2 and 3 are listed with the stratigraphy in Table 1.

In summary, all the lavas have reversed (R) polarity, the bulk of the Cuillin [except for some late units, the Coire Uaigneach Granophyre (CU) and some basaltic cone sheets] and Srath na Creitheach Centres are also R, the Western Red Hills Centre is normally magnetized (N) [except for the Loch Ainort Epigranite (LAE) which is R], while the Eastern Red Hills Centre seems to be entirely R. Dykes of both polarities occur, as would be expected if intrusion occurred throughout the activity, with R predominating [19 per cent are N; Wilson *et al.* (1974, 1982)]. This simple picture, where (almost) all units within a centre have the same polarity, is consistent with the fact that the units within a centre share a common focus and so were probably emplaced during a relatively short time span with possible breaks in activity between centres. The weighted mean directions for the polarity groups within each centre are well defined and support this idea, e.g.:

WRHC, before LAE; $D = 358°$, $I = 76°$, $\alpha_{95} = 6°$; N
WRHC, after LAE; $D = 356°$, $I = 80°$, $\alpha_{95} = 3°$; N
WRHC, all except LAE; $D = 357°$, $I = 79°$, $\alpha_{95} = 2°$; N
but
CC, except normal sites; $D = 188°$, $I = -68°$, $\alpha_{95} = 2°$; R
compared with
ERHC; $D = 187°$, $I = -62°$, $\alpha_{95} = 3°$; *R*.

As will be shown later (see Discussion), the activity spanned at least the period 59.3 ± 0.7 to 53.5 ± 0.4 Ma and involved at least five polarity intervals, R–N–R–N–R, with the possibility, depending on the position of the LAE within the sequence, that there may be seven.

However, this summary conceals some inconsistencies within events and within centres. These problems and other details of the results are described below, and in the Discussion suggestions are made on how they can be reconciled with the geological sequence and the polarity time-scale.

(a) Beinn Totaig group lavas

This group was not included in the earlier work (Wilson *et al.* 1972a) which found that lavas of the other four groups were all reversely magnetized but with a spread of directions.

Five of the six flows sampled gave S–SW directions of magnetization with negative dips which extend the grouping of directions noticed by Wilson *et al.* (1972a; Fig. 3). Two fall into our intermediate classification (Wilson *et al.* 1972b) but the overall mean direction is reversed and lies close to the direction of the Bracadale set (Wilson *et al.* 1972a). Thus, all five groups of lavas in Skye, like those elsewhere in the British Tertiary igneous province (BTIP), are reversely magnetized.

(b) Ring Eucrite and Outer Gabbro (event 2, Cuillin Centre)

Site U111 in the Gars-bheinn Gabbro and sites in Coire na Creiche give reversed, though not identical, directions. However, site U110 in the Gars-bheinn Gabbro and a site (U090) in Glen Sligachan are normally magnetized and demagnetization provides no evidence for partial overprinting.

Samples of U110 show petrographic evidence for alteration, the plagioclase is very cloudy and the pyroxene intensely dark due to the presence of magnetite, so the site could have been magnetically overprinted but, as it is surrounded by reversely magnetized rocks, by what agent? As there is some uncertainty in the field evidence as to which event this gabbro really belongs (Weedon 1961; Bevan & Hutchison 1984), it is possible that this site belongs later in the succession.

Site U090 is located in a wedge-shaped gabbroic mass (Fig. 3) between two intrusions of the later Western Red Hills (WRHC), the Marsco and Glamaig Epigranites, both of which are normally magnetized. The site is less than 100 m from these later intrusions and is highly altered showing brecciation and the extensive development of secondary alteration minerals, including chlorite, amphibole and calcite, and with the clinopyroxene very cloudy due to the exsolved magnetite which in some crystals is in the form of numerous closely spaced rods. It is likely, therefore, that site U090 has been completely thermally overprinted by the WRHC (see Discussion); the direction obtained is close, though not identical, to that of U028 in the nearby Marso Epigranite.

(c) Border group (event 5, CC)

Of the two sites sampled, both are R.

(d) Eucrite series (event 8, CC) and Cone Sheets (event 17, CC)

Of the 16 sites in the Eucrite, all are R except U065 which is N, and U068 which has intermediate direction (see Fig. 3). These apparent anomalous directions are probably due to thermal overprinting by the Western Red Hills Centre to the north. U065, on the north side of Garbh Bheinn, is only *circa* 100 m from the contact, while U068, which is the next closest to the Western Red Hills Centre at *circa* 250 m, shows clear evidence, during demagnetization of N polarity superimposed on R (Fig. 5b). An alternative, or additional, cause of overprinting could have been the intrusion, in the vicinity of U065, of cone sheets of event 17 (U066 and U067) at a time when the field was normal. However, all four sites, U065-8, show petrographic evidence of metamorphism, with U068 the least affected. Plagioclase is clouded and the pyroxene has exsolved opaque iron oxide which is often more dense in the marginal parts of the

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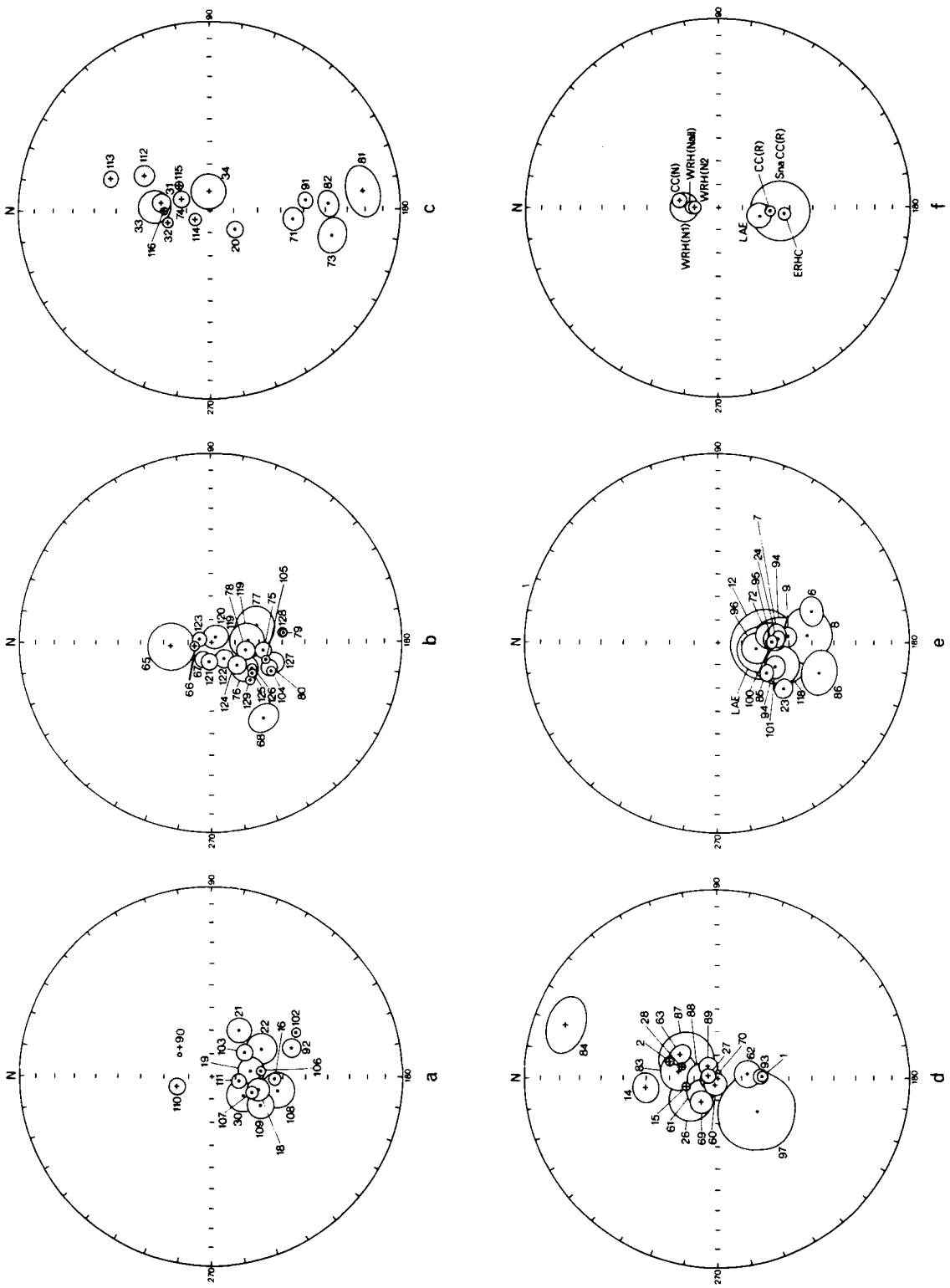


Figure 4. Equal-area stereographic plots of palaeomagnetic directions; (+ positive dips, ● negative dips). (a) Cuillin Centre (events 2, 4, 5, 7, 9 and 13 of Table 1). (b) Cuillin Centre, Eucrite series and cone sheets cutting gabbro [events 8 and 17(ii)]. (c) Cuillin Centre, Coire Uaigneich and other granites with intersecting cone sheets and a dyke [events 16 and 17(i)]; Srath na Creitheach Centre (events 20 and 21). (d) Western Red Hills (events 26–34, 36 and 37). (e) Eastern Red Hills (events 39–48) and late dykes. Unweighted mean direction of four sites from the Loch Ainort Epigranite is included for comparison. (f) Weighted mean directions for each centre and the Loch Ainort Epigranite.

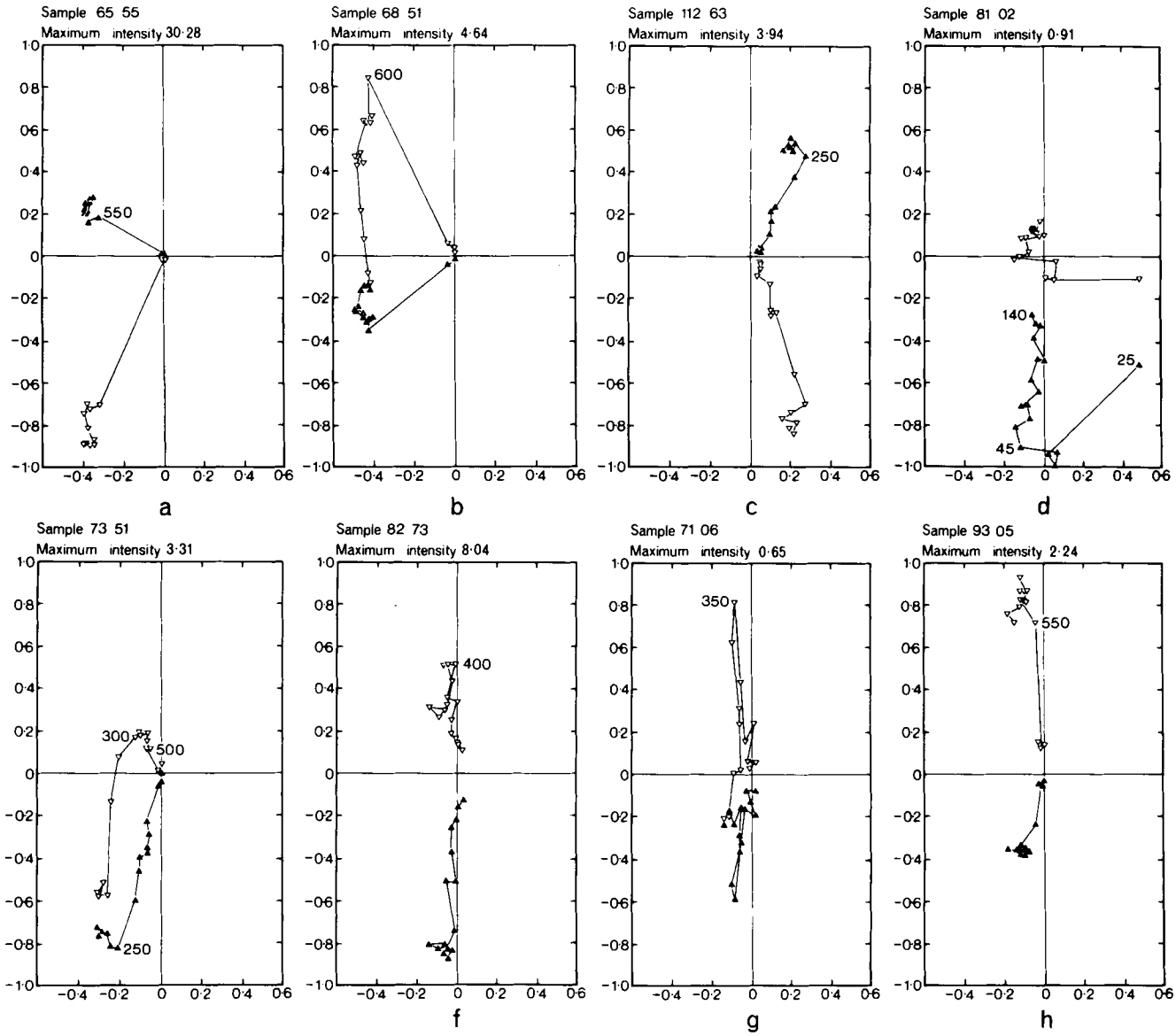


Figure 5. Composite vector diagram for thermal or alternating field (81–02) demagnetization of selected samples. Vertical axis, N and up; horizontal axis, E, ▽ vertical plane; ▲ horizontal plane. Numbers are temperatures (°C) except for sample 81–02 where they are peak alternating fields (milliTesla). Intensity of magnetization in units of 10^{-5} Am^2 kg^{-1}.

crystals. Some pyroxene crystals are partially replaced by secondary green amphibole and olivine has been pseudomorphed by talc, magnetite and chlorite. In the same area, but further away from the contact, sites in the gabbro (U119-122), and a cone sheet (U123) give R so it is likely that overprinting of both Eucrite and some of the cone sheets due to the proximity of the Western Red Hills Centre (see also Discussion) has occurred.

(e) Coire Uaigneich (CU) and other Granites (event 16, CC) and cone sheets (event 17, CC)

These bodies, which are considered to be late members of the Cuillin Centre, were sampled in three localities revealing a mixture of N and R polarities.

In Coire Uaigneich itself (Bell 1976, fig. 5) the granophyre (four sites; U031, U112, U114, U115) together with cone sheets cutting it (four sites; U032, U033, U113, U116) and a dyke (U034, event 18?) cutting one of the cone sheets are all N. There is no suggestion of overprinting from the demagnetization data nor is there an adjacent normally magnetized body.

Secondary epidote was found in the CU but only as veins in what was otherwise a remarkably fresh cone sheet (U116). In fact the cone sheets cutting the granophyre in Coire Uaigneach are significantly fresher than the granophyre. Ferry (1985b) states that the temperature range over which hydrothermal alteration caused most mineralogical changes in the granites is 350°–450 °C. If these temperatures are correct it is unlikely that the CU was reheated to a temperature high enough for complete magnetic overprinting to occur.

At Camasunary, samples from the granophyre at site U074 are N, while those from site U081 are intermediate (southerly, +ve dip) with a trend during demagnetization implying that a primary R vector has been overprinted

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(Fig. 5d). Petrographic examination reveals that some pyroxene crystals are remarkably fresh. This rock has not suffered extensive alteration which could have been produced either by the later dolerite cone sheets or by the pervasive hydrothermal system associated with the centre. In this area, too, cone sheets assigned to event 17 are observed to cut the granophyre. U073 (near site U074) has a shallow reversed direction and thermal demagnetization shows it had a normal overprint (Fig. 5e), though this may be simply low-temperature viscous remagnetization as the plagioclase and pyroxene are very fresh. Cone sheet U082 (near site U081) is reversed (Fig. 5f). Other cone sheets nearby (U076, U078, U080) are also reversed.

Samples were collected from a site, U071 (Fig. 3), on the SW slopes of Belig with the understanding that it formed part of the Coire Uaigneach Granophyre (Bell 1976, Fig. 5). The samples, however, are petrographically different from the Coire Uaigneach Granophyre *per se*, though the epigranite may be part of a suite which includes the granophyre (Bell 1966, pp. 340–1). Magnetically, this site gives R polarity though demagnetization shows it to have acquired a secondary normal component (Fig. 5g). It is too far from the reversely magnetized ERHC to have been overprinted by it (see Discussion); there is the possibility that this intrusion belongs to the ERHC rather than the Cuillin Centre (J. D. Bell, personal communication).

Our tentative conclusions are that the granite/granophyre at Coire Uaigneich and Camasunary are related units of the Cuillin Centre and were formed in an N interval, while the contrast in both petrography and magnetization show that the mass at Belig is a separate intrusion; either an earlier one of the Cuillin Centre when the polarity was R or a later one of the ERHC. The cone sheets assigned to event 17 contain both polarities and so either the event straddled an inversion of the magnetic field or was made up of two discrete episodes. Since one of the cone sheets in Coire Uaigneich is cut by a normally magnetized dyke assigned to event 18, two episodes is more probable.

The Coire Uaigneach Granophyre, normally magnetized and dated at 59.3 ± 0.7 Ma (Dickin & Exley 1981), is a very significant intrusion in determining the early part of the magnetic polarity sequence.

(f) Loch Ainort Epigranite (LAE) (event 30, WRHC)

The whole of the Western Red Hills Centre has N polarity apart from this unit which is reversely magnetized (and one of the four sites in the Marscoite Suite which is I). The LAE is believed to have been intruded mid-way through the WRHC sequence, so that it has the opposite polarity to the intrusions emplaced immediately before and after it. To check the validity of this R polarity, the LAE was sampled at four widely separated sites (Fig. 2) all of which give a clear reversed polarity. The mean of the four site mean directions each accorded unit weight (unweighted unit mean) is $D = 191.2°$, $I = -72.6°$, $\alpha_{95} = 9°$.

A dyke, U002, assigned to event 37, which cuts the LAE is N.

Possible explanations for the polarity of the LAE are as follows.

(1) It has been remagnetized. We consider this unlikely because of the distance from the sites of the nearest R body of the Eastern Red Hills Centre; though the margin of the centre is concealed by the waters of Loch Ainort, site U093 is at least 1.5 km from the contact and the evidence is against overprinting to this distance (Fig. 5; see Discussion). Moreover, overprinting would need to be selective to leave sites in the Northern Porphyritic Felsite (NPF) (U026), Southern Porphyritic Epigranite (SPE) (U083), Marscoite Suite (MS) (U074) and the dyke (U002) unaffected.

(2) It is a later intrusion than shown in Table 1 and belongs to the ERHC. Evidence for its position within the WRHC sequence is as follows: it is petrologically similar to the Bheinn Dearg Mhor Epigranite (BDME), and is separated from it by a crush zone, leading Bell (1966, p. 322) to the view that the LAE might be a segment of the BDME elevated by faulting. Subsequently Bell (1976, table 1, p. 256) listed them separately and we have adopted his sequence in Table 1. Further large-scale cross-cutting relationships suggest that the LAE is older than the Southern Porphyritic Epigranite (SPE) (B. R. Bell, private communication). This position is also supported by the fact that dyke U002, which probably belongs to the closing event (37) of the WRHC, is normally magnetized. This polarity is unlikely to be the result of overprinting because all known later bodies sampled (ERHC and later dykes) are reversely magnetized though the intrusion of even younger N dykes cannot be ruled out.

On the other hand, the field evidence is not unequivocal and Bell (1966) also states that the LAE is finer grained towards the SPE, suggesting chilling, which would require the LAE to be the younger of the two. If the LAE was intruded later than hitherto thought, then it is probably part of the ERHC to which it is contiguous; in this case the normal dyke, U002, cutting it would have to be very late in the succession [but see **(h)**]. However, the confidence limits of the weighted unit mean direction of the LAE ($D = 193.0°$, $I = -72.5°$, $\alpha_{95} = 5.2°$), and the ERHC group mean ($D = 186.7°$, $I = -61.7°$, $\alpha_{95} = 2.8°$) do not overlap. The mean direction lies closer to that for the CC ($D = 188.0°$, $I = 68.3°$, $\alpha_{95} = 2.3°$) but the proposition that the LAE is earlier is considered improbable (B. R. Bell, private communication).

The radiometric dating evidence is also equivocal (see following section.)

(3) It was formed during a brief interval of R polarity. If the geological sequence is correct and there has been no overprinting there must have been a short interval of R polarity during the longer interval in which the bulk of the WRHC was emplaced. In this case the intrusion of the LAE was probably preceded and followed by a significant pause in activity.

At present there is no totally convincing geological field evidence that the geological sequence (Table 1) is incorrect so that (3) would appear to be the best interpretation. However, a time break seems more likely between centres than within a centre and given the debatable nature of the field evidence, the simpler reversal pattern offered by (2) is appealing.

(g) Beinn na Cro (BNCE[1]), Allt Fearna (AFE) Granites and Beinn an Dubhaich Epigranite (BADE) (events 42, 43 and 44, ERHC) and Creag Strollamus Epigranite (CSE) (event 49)

Stewart (1965) suggested that the first three of these bodies may be part of the same arcuate mass. Unfortunately, their

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remanent magnetism is weak and unstable, with $Q < 0.5$, (because the rocks are almost devoid of ferro-magnesium minerals, and in U011 the iron oxides have been replaced by secondary minerals including haematite) and supports Stewart's suggestion only to the extent that the three granites are similarly poor magnetic material.

Bell & Harris (1986, p. 91) have found the Creag Strollamus Epigranite to be compositionally similar to the other three and, based on its structural relationships, conclude that it, too, is part of the same intrusive mass. Its palaeomagnetic results are also inconclusive.

(h) Dykes

The regional dykes are of both polarities and include a few examples of R dykes cutting N ones and vice versa, which requires a minimum polarity sequence of R–N–R or N–R–N (Wilson *et al.* 1974, 1982).

The two dykes (U095 and 96) that cut the composite intrusion U094 (event 48) both give R polarity and, together with dykes U024 (cutting the Beinn na Caillich Granite, event 47), U100 and U101 (cutting the Beinn an Dubhaich Granite, event 44) suggest that the last phase of dyke intrusion (also the last phase of igneous activity in Skye) occurred during a reversed polarity interval. It is therefore unlikely that the normally magnetized dyke, U002, cutting the Loch Ainort Epigranite is as young.

RADIOMETRIC DATING AND THE SPAN OF ACTIVITY

As the only reliable radiometric dates for Skye have been obtained by the Rb–Sr isochron method, which is effectively confined to the acid rocks, 15 samples were analysed by the ^{40}Ar–^{39}Ar step-heating method. Of these, only a few came near to satisfying the plateau acceptance criteria of Lanphere & Dalrymple (1978), viz. a plateau should comprise at least four steps and half the released Ar, and on a correlation diagram the selected plateau steps should fit a straight line with a mean square weighted deviate (MSWD) less than 2.5 (Brooks, Hart & Wendt 1972), with the line giving an age consistent with the plateau age and an argon contamination ratio not distinguishable from that of air argon, 295.5.

Analyses followed the procedure described in Mussett (1986), and the results are summarized in Table 2. Only those that constrain the period of activity will be considered further.

Since neither of the two lava samples analysed yielded a plateau the age of the lavas is not known directly. However, erosional debris intercalated with the Skye lavas show that these lavas were probably contemporaneous with the NW lavas of Rhum (Meighan *et al.* 1982), which have been dated as 60.4 ± 1.0 Ma (Mussett 1984). [All dates have been calculated using the constants recommended by Steiger & Jäger (1977) and errors are quoted at the 1σ level].

A basic minor intrusion in the western part of the Cuillins Centre (U021, analysis R1469) yielded a 20-step age spectrum (Fig. 6a). Steps 5–18 (step 10 has been omitted because it suffered technical difficulties) give a correlation plot with MSWD of 2.7 and argon contamination ratio significantly above 295.5, and so fail the criteria but the weighted mean of these steps, 61.4 ± 1.5 Ma, suggests an age for the early part of the Cuillins of 60–61 Ma.

This tentative date is consistent with the oldest reliable date obtained for the Cuillins Centre, an Rb–Sr isochron date of 59.3 ± 0.7 Ma for the Coire Uaigneich Granophyre (Dickin & Exley 1981), the youngest major intrusion of the Cuillins. Though the Rb–Sr samples came from the Camasunary exposure of this granophyre, where the polarity is less clear, this date is taken to be in the oldest normal polarity interval recorded by the Skye Tertiary igneous succession.

There are no analyses for the succeeding and small Srath na Creitheach Centre, and no fully reliable ones for the following Western Red Hills Centre. Dickin (1981) analysed samples of the Loch Ainort Epigranite but his Rb–Sr isochron had a high MSWD of 4.8. If the scatter is due only to random variation in the initial strontium isotope ratio then the data yields a date of 58.7 ± 0.9 Ma. A sample of the Beinn Dearg Mhor Epigranite failed to give a ^{40}Ar–^{39}Ar plateau.

There are several dates for the last centre, the Eastern Red Hills Centre. Dickin (1981) obtained a satisfactory Rb–Sr isochron date of 53.5 ± 0.8 Ma for the Beinn an Dubhaich Granite. A ^{40}Ar–^{39}Ar analysis of the same intrusion gave a generally saddle-shaped age spectrum (Fig. 6b), so that the short 'plateau', 55.8 ± 1.4 Ma is taken as a maximum age estimate, is consistent with the Rb–Sr date. A sample, RE47, of the main felsite part of the Rudh' an Eireannaich Sill, one of the late composite and acid sills, gave a good plateau (Fig. 6c), with a date of 55.3 ± 0.9 Ma, and another sample of the same sill, U094B, though it gave a somewhat erratic spectrum (Fig. 6d) also satisfied the criteria and gave a similar date, 55.0 ± 0.8 Ma. A sample of basified granophyre from the Kilchrist Hybrid Ring Dyke gave a short plateau of 54.0 ± 1.4 Ma (Fig. 6e).

In summary, the available radiometric dates do not cover the span of Tertiary activity as well as one would like but they show that activity occurred over at least the approximate interval 61–54 Ma. According to the polarity time-scales of both Harland *et al.* (1982) and Berggren *et al.* (1985) such a period must have spanned (parts of) at least five polarity intervals. For comparison, the comparable-sized succession of Mull lasted only about 3 Ma (Mussett 1986) and occurred during only three polarity intervals, R–N–R (Dagley, Mussett & Skelhorn 1987). If the date of the Loch Ainort Epigranite is correct it indicates that activity did not occur evenly over the period, which raises problems of how to match the units of the Skye succession to the polarity time-scale.

DISCUSSION

This section uses the available data to address several problems.

(i) Tectonic effects

It is quite probable that within a central intrusive complex, because of the mode of emplacement involving ring faulting, earlier intrusions may have been deformed, tilted or displaced due to the emplacement of younger intrusions. For the vast majority of cases this is very difficult, if not impossible, to prove unless one can map structures in an

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Table 2. Radiometric dating analyses.

Unit	Sample No	Analysis No	Total Fusion age, Ma	Best 'plateau' Steps	Best 'plateau' %39 Ar released	Best 'plateau' Age, Ma [1]	Correlation plot Age, Ma	Correlation plot Contamination Argon ratio	MSWD	Comment
Eastern Red Hills Centre										
Composite Sill (acid component	U094B	R1633	72	7-11	52	55.0 ± 0.8	57.0 ± 1.0	281 ± 11	1.4	satisfies criteria
" " (Rudh 'an Eireannaich)	RE47	R1529	58	2-8	65	55.3 ± 0.9	55.1 ± 0.7	298.9 ± 8.8	1.9	satisfies criteria
Kilchrist Hybrid Ring Dyke	80/1	R1359	55	3-8	45	54.0 ± 1.4	53.9 ± 1.4	298.5 ± 6.5	2.2	short plateau
Beinn an Dubhaich Granite	U006	R1398	58	8-13	44	55.8 ± 1.4	55.6 ± 1.5	300 ± 17	0.9	see text
Broadford Gabbro	U012	R1350	59	9-21	78	58.6 ± 1.4	56.2 ± 0.5	319.5 ± 5.5	2.3	non-atmospheric contaminant ratio
Western Red Hills Centre										
Beinn Dearg Mhor Epigranite	U013	R1353	66	none	-	-	-	-	-	fails criteria
Cuillins Centre										
Druim nan Ramh Ring Eucrite	U016A	R1351	90	none	-	-	-	-	-	fails criteria
Minor intrusion	U021	R1469	63	5-9,11-18	79	61.4 ± 1.5	60.6 ± 1.3	308.3 ± 5.0	2.7	see text
Boulder	U021Y	R1468	88	11-13	59	72.2 ± 1.6	128 ± 22	390 ± 215	1.0	fails criteria
Lavas										
N sequence [2]	N026	R1631	31	6,8,9	70	60.2 ± 0.7	58.7 ± 0.7	300.9 ± 1.7	0.01	non-atmospheric contaminant ratio
"	N001	R1632	52	6-14	76	59.5 ± 2.3	62.0 ± 2.9	276 ± 36	23	large MSWD
Others										
Dyke cut by Beinn an Dubhaich Granite	U007	R1348	130	none	-	-	-	-	-	fails criteria

Errors are ±1 σ throughout the table
Dates are calculated using the constants recommended by Steiger and Jäger, 1977
[1] Average of step-ages weighted by the inverse of their variance
[2] Wilson et al. 1972a

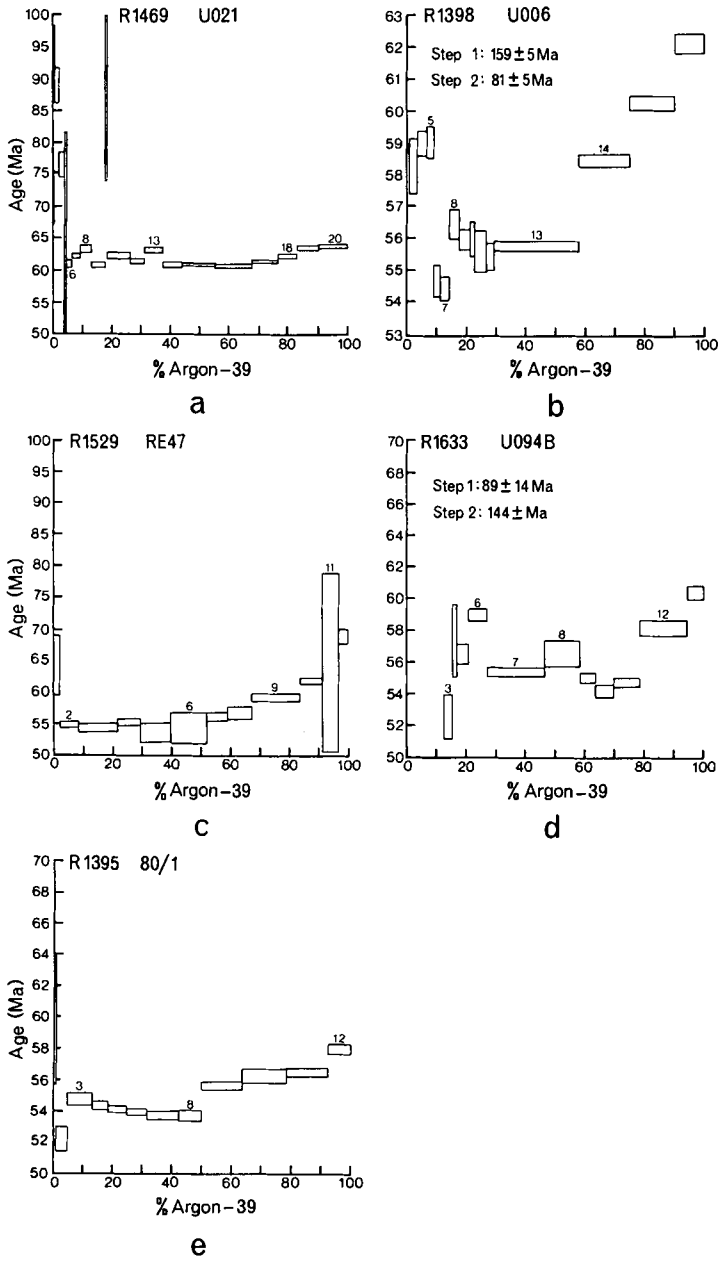


Figure 6. Radiometric age spectra of selected samples. The rectangles give data for the analysis of the gas released in steps of progressively higher temperatures: the horizontal length denotes the fraction of ^{39}Ar within the sample released in that step, while the vertical extent reflects the apparent age at the $\pm 1\sigma$ error level; the small numbers above some of the boxes are step numbers. Table 2 provides additional information.

earlier intrusion which can be shown to have been deformed by a later intrusion. Even then, the deformations may have taken place while the rocks were above the Curie temperature.

Gabbro U90 (event 2) is brecciated which could indicate movement resulting from the emplacement of intrusions belonging to the WHRC. U92 (Cuillin Gabbro, Glen Sligachan) is located within a lenticular shaped screen of older country rock separating two later intrusions and may well have been displaced from its original position. However, tilting does not appear to be important in either case, the normal polarity of U90 almost certainly resulting from overprinting.

At places within the Cuillin Complex the layering dips at high angles; for instance, Weedon (1965) records dips of 35°–80° for the Sgurr Dubh ultrabasic mass (event 7). If, as believed by Weedon, the layering was produced by bottom accumulation of crystals on successive, more or less, horizontal floors then the high dips of the layering are due to later movement, perhaps related to a ring fault.

The situation is similar to that of the Hypersthene Gabbro of the Ardnamurchan Igneous Complex (Dagley *et al.* 1984) and as in that case the magnetic evidence, though less complete, indicates that the high dips of the layering were acquired when the rocks were above the Curie temperature.

Tilting of the Plateau lavas has been discussed by Wilson

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et al. (1972) who concluded that it pre-dated the magnetization.

(ii) Remagnetization

The evidence from demagnetization and auxiliary experiments suggests that, except for the few cases mentioned, a stable primary remanence has been isolated. However in a region such as this, affected by hydrothermal alteration, it is possible that secondary chemical or thermal magnetizations have been acquired which mask the primary remanence. The results of petrographic examination have been described above for a number of cases of suspected remagnetization. In addition to the mineralogical changes outlined in the Introduction, magnetically overprinted basic rocks exhibit clouding of the plagioclase, and the pyroxene is turbid due to exsolved magnetite, frequently in the form of rods, which are often more densely developed at the margins of the crystals. The same changes were noted by Dagley *et al.* (1987) in the magnetically overprinted basic igneous rocks of Mull.

Even in the absence of a hydrothermal system, there is the possibility with any succession of contiguous intrusions that earlier intrusions have been reheated and remagnetized by later intrusions or, alternatively, that a considerable part of the complex cooled so slowly that the magnetic blocking temperatures were reached only a significant time after its intrusion. The geometry of the central igneous intrusive complex allows the extent of any such overprinting to be estimated.

There is clear evidence that the WRHC has caused remagnetization of the Eucrite Series of the CC [see Palaeomagnetic Results, paragraph **(d)** above]. Total overprinting apparently occurred at site U065, partial overprinting at U068 and little or no overprinting at sites U119–121, respectively *circa* 100, 250 and 300 m from the contact. Evidence that overprinting does not extend far also comes from site U016 in the CC which is R and unaffected though only *circa* 700 m from the WRHC. Thus the WRHC seems to have been able to produce overprinting to only about 100–200 m. This limit is consistent with results for Mull (Dagley *et al.* 1987) where the zone of overprinting was shown to be less than 600 m wide.

These distances make it very unlikely that the N polarity of site U110 and the CU or the R polarity of site U071 on the slopes of Belig and the LAE are due to overprinting by later intrusions, as judged at the present level of erosion. Conversely, they make it probable that the N polarity of site U090, part of event 2 of the Cuillins Centre and almost enclosed by the Western Red Hills Centre, has been totally overprinted.

Independently of overprinting by major units or centres, overprinting can also be produced by minor intrusions and though this will be very local, the intrusion could escape detection if the exposure is unfavourable.

Storetvedt (1978) has claimed that the Skye lavas have been remagnetized as a result of chemical alteration over a long period in the Middle-to-Late Tertiary and that 'the original TRM only plays an insignificant role in the present magnetization of the investigated Skye lavas.' He bases this conclusion on the existence of shallow magnetic dips found in many samples, which he believes reflects a mid-Tertiary pole position, and on the complex behaviour of some samples in various combinations of alternating field and thermal demagnetization.

It is difficult to reconcile his conclusions with the field evidence. For instance, thin sequences of lavas in northern Skye have closely the same direction which can be correlated laterally over a distance of about 7 km (Khan 1960). Further, the dykes which cut the lavas contain both R and N polarities. Yet another argument against remagnetization is provided by the Sleat dyke swarm in which normal and reversed directions are antiparallel to within 0.7° (Wilson *et al.* 1982). In the central intrusive complex the correlation of polarity with geological units (except where thermal overprinting has probably occurred) also weighs against Storetvedt's conclusion.

It is concluded that though chemical remagnetization may have produced some scatter of directions, unlike the thermal remagnetizations discussed above it has not affected the polarities of samples.

(iii) Absence of normally magnetized lavas

A remarkable feature of the BTIP is that though all the lavas are reversely magnetized [for a review see Mussett, Dagley & Skelhorn (1988)], most of the regional dyke swarms contain a proportion of normally magnetized dykes. Did the normal dykes feed normal lavas and, if so, where are the normal lavas?

In Skye there is some geochemical evidence pertinent to this problem. Mattey *et al.* (1977) have shown that there were three compositionally distinct basaltic magma types which they have called the Skye Main Lava Series type, the Preshal Mhor type and the Fairy Bridge type. These three types appear to be distributed very unequally among the lavas and dykes, for nearly all of the lavas belong to the Skye Main Lava Series type, whereas only about 15 per cent of the dykes do so; 70 per cent of the dykes are of the Prehal Mhor type which is found only rarely among the lavas; while the remaining 15 per cent of the dykes are of the Fairy Bridge type of which only one lava is known. Thus many of the dykes are of compositions occurring only rarely among the lavas and so it was thought possible that the few lavas of these compositions might include normally magnetized ones.

Furthermore, Mattey *et al.* (1977) consider that the Preshal Mhor magma type came relatively late in the Skye igneous sequence and that dykes of this type probably were associated with the emplacement of the central complex, and some are known to cut the granites (e.g. U002). Also, the lavas at Preshal Mhor itself and the nearby Preshal Beg occur high in the existing lava sequence, having been extruded into valleys cut in the Osdale Group. Evidence that the Preshal Mhor type dykes were not confined to one polarity period in the development of the Skye Tertiary rocks, and so may include late dykes, comes from the observation that though these dykes are concentrated towards the axis of the main dyke swarm there is no correlation of the percentage of normally magnetized dykes with distance from the axis (R. L. Wilson, personal communication). Of the two sampled dykes known to be of Preshal Mhor type, one in the Western Red Hills (U002), is

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normally magnetized, while one in the Eastern Red Hills (U007) is reversed (Table 1).

Notwithstanding these arguments for the likely occurrence of N lavas, none of the sampled lavas of Preshal Mhor or Fairy Bridge type is normally magnetized. Might this imply that there were no normally magnetized lavas? This is difficult to believe. Surely, one or two dykes of the N dykes reached the surface and fed lavas. Also, some of the N intrusions within the central intrusive complex may well have vented onto the surface, feeding lavas. One possibility is that the N dykes were relatively late and so high in the sequence and that the main reason why no N lavas have been found is because of the dpeth of erosion. The distribution of zeolites in the Mull lavas suggests that 1000 m of lavas have been eroded from the Mull pile.

(iv) The polarity sequence

Taken at face value the magnetic results indicate that all the lavas and most of the Cuillins Centre are R, the late units of the Cuillins Centre, including the Coire Uaigneich Granophyre (16, 17 and 18) are N, the following Srath na Creiteach Centre is R, the Western Red Hills Centre (apart from the Loch Ainort Epigranite) is N, while the Eastern Red Hills Centre is again R, giving a sequence R–N–R–N–R. If the relative age of the Loch Ainort Epigranite as given by Bell (1976) is correct, then the igneous activity spans seven polarity intervals, R–N–R–N–R–N–R. However this seems to require a short R polarity interval during which only the Loch Ainort Epigranite formed, and the only short R interval so far proposed is

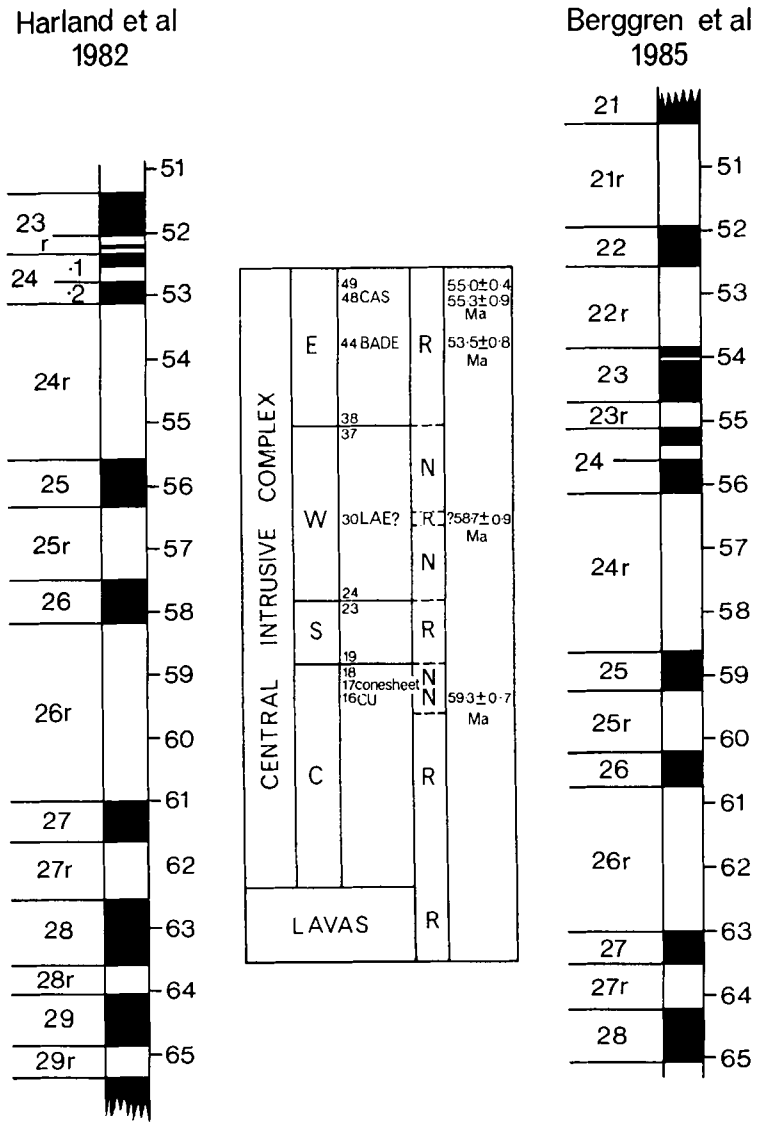


Figure 7. Tentative magnetic polarity stratigraphy of the Skye Tertiary igneous rocks, with two polarity time-scales, one on each side, shown for comparison. The height of the stratigraphic column indicates the age range involved, but the position of the events is only approximate because of uncertainties involved so that the polarities associated with the stratigraphy are not identified with any particular chrons of the time-scale. The four centres of the intrusive complex are denoted by C (Cuillin), S (Srath na Creitheach), W (Western Red Hills) and E (Eastern Red Hills). The numbers in the central column mark the bounding events, the events having a polarity contrasting with the centre as a whole and the dated events. The measured polarities are shown in the adjacent column and the radiometric ages in the right-hand column.

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within chron 24. On the Harland *et al.* (1982) time-scale this is at *circa* 52.7 Ma and on the Berggren *et al.* (1985) time-scale, at 55.5 Ma (Fig. 7), both of which are incompatible with the radiometric date for the LAE (58.7 ± 0.9 Ma, which admittedly is of low reliability). It is of course conceivable that the LAE is the only unit carrying a record of one of the longer reversed intervals with long breaks in the activity both before and after its intrusion.

When it comes to matching intervals to specific polarity chrons, the Harland *et al.* (1982) scale, with its long 26r chron would more easily encompass the R lavas and most of the Cuillins Centre, plus the earlier rocks of Rhum and Eigg which also are R and probably formed in the same interval (Mussett 1984; Mussett *et al.* 1988), but requires the date of the normally magnetized Coire Uaigneich Granophyre to be near its younger limit to fit into chron 26. This time-scale would accommodate the rest of the sequence with the exception of a short R interval for the Loch Ainort Epigranite. The time-scale of Berggren *et al.* (1985) also presents problems for it would require all the lavas and most of the Cuillins Centre to have been formed in chron 25r which is only *circa* 1 Ma long, while the small Srath na Creiteach Centre would be allocated to the long chron 24r, displacing the R interval of the Loch Ainort Epigranite into the very short R interval within chron 24.

The fact that the boundaries between most or all of the four centres coincide with a change of polarity suggests that there were significant intervals between them.

It may also be noted that the young ages for the late members of the ERHC suggest that they were contemporary with the granites of Lundy (54.8 ± 1.4 Ma; Miller & Fitch 1962; Dodson and Long 1962; 53 ± 5 Ma; Hampton & Taylor 1983); the Mourne Mountains (56.0 ± 0.7 to 53.3 ± 0.6 Ma; Gibson *et al.* 1987; Thompson, Mussett & Dagley 1987); St. Kilda (55 ± 0.5 Ma; Brook 1984) and Rockall (52 ± 8 Ma; Hawkes *et al.* 1975).

CONCLUSIONS

(i) The limited number of reliable radiometric dates available show that the Tertiary igneous activity began some time before 59.3 ± 0.7 (1σ) Ma and continued until after 53.5 ± 0.8 Ma.

(ii) The lavas and most of the Cuillins Centre were formed during an R interval with the three last events [CU(16), cone sheets (17), dykes (18)] of the Cuillins Centre formed during an N interval. The ensuring Srath na Creitheach Centre formed during an R interval, then the following Western Red Hills Centre (apart from the Loch Ainort Epigranite) in an N interval and the last centre, the Eastern Red Hills, in an R interval. This gives the polarity sequence R–N–R–N–R. Even if additional polarity intervals are found to be necessary it is very probable that activity began and ended when the magnetic field was reversed. The reversely magnetized dykes which cut the Beinn an Dubaich Epigranite and Composite and Acid Sill are probably the last manifestation of igneous activity in Skye.

(iii) The Loch Ainort Epigranite, the only R body in the Western Red Hills Centre, was either formed during a short R interval, making the sequence R–N–R–N–R–N–R, or has been assigned an incorrect position in the sequence, in which case it is probably part of the Eastern Red Hills. Neither alternative is without difficulties.

(iv) It is difficult to identify the observed polarities with specific chrons of polarity time-scales of either Harland *et al.* (1982) or Berggren *et al.* (1985). In particular, a short interval to accommodate the Loch Ainort Epigranite presents problems.

(v) There was a change of polarity between the formation of most or all of the centres. This suggests that there were significant intervals between their formation.

(vi) The bodies mapped as belonging to event 16, Coire Uaigneich and other Granites, yield N and R polarities. The evidence indicates that the bodies in Coire Uaigneich and at Camasunary (N polarity) are correctly assigned in the sequence but the body at Belig (R polarity) is not contemporary and may be an earlier member of the Cuillins Centre or belong to the Eastern Red Hills Centre. The cone sheets assigned to event 17 contain both N and R members, and either their intrusion spanned an inversion of the magnetic field or they belong to two episodes.

(vii) Overprinting by major intrusions or centres occurs but total overprinting, producing complete remagnetization, only extends to about 200 m.

(viii) No normally magnetized lavas have been found in any of the five structural groups though normally magnetized dykes exist. A probable reason is that the flows associated with normally magnetized dykes were relatively late in the sequence and have been removed by erosion.

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